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Effects of shoe collar height and limb dominance on landing knee biomechanics in female collegiate volleyball players

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ABSTRACT

Volleyball-specific footwear with higher collar heights (a mid-cut shoe) are worn to restrict ankle motion. Reduced ankle dorsiflexion has been associated with increased frontal plane motion and injury risk at the knee. With the high frequency of unilateral landings in volleyball, the purpose of this study was to determine the effect of volleyball-specific shoes and limb dominance on knee landing mechanics in collegiate volleyball players. It was hypothesized that participants would exhibit smaller sagittal plane and greater frontal plane knee joint mechanics in mid-cut and dominant limb and that vertical and posterior directed ground reaction forces would be greater wearing mid-cut, yet similar between limbs. Seventeen female volleyball players performed unilateral landings on each limb in mid-cut and low-top volleyball shoes. For shoe main effects, smaller peak dorsiflexion angle and internal peak plantarflexion moment and greater peak medial ground reaction force were found in the mid-cut but with no impact on knee mechanics. For limb main effects, the internal peak knee abduction moment was greater in the dominant limb. Greater peak lateral ground reaction force was found in the interaction between the non-dominant limb and low-top. Further research is warranted to better understand shoe and limb impact in volleyball players.

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Volleyball; shoe collar;
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1. Introduction

Female participation in volleyball has expanded to reach record high participation rates at the high school and collegiate level in the USA (Chandran et al., 2021; Tillman, Criss, et al., 2004; Tillman, Hass, et al., 2004). Volleyball is an explosive sport that requires the athletes to perform repetitious bouts of jumping and landing (Bisseling et al., 2008). Sixty-three percent of musculoskeletal injuries sustained in volleyball occur during the jumping and landing tasks of spiking and blocking, and over half of the injuries reported are non-contact or overuse injuries (Chandran et al., 2021; Xu et al., 2020). For female college volleyball players, knee injuries are the most prevalent injury reported, with anterior cruciate ligament (ACL) ruptures being the most common injury to the knee (Chandran et al., 2021; Ferretti et al., 1992). Two risk factors of ACL injuries important to address in a volleyball population: a restriction in ankle dorsiflexion range of motion and limb dominance (Fong et al., 2011a; Ford et al., 2003a; Schroeder et al., 2021).

The expansion of volleyball as a sport has been accompanied by the growth of the footwear industry and design of volleyball shoes. The design of volleyball shoes is relevant to the ACL risk factor of restriction in ankle dorsiflexion range of motion as shoes with different levels of collar height have been demonstrated to alter ankle range of motion (Cronin, 2001; Daack & Senchina, 2014a; Lam et al., 2015). Anecdotally, collegiate volleyball coaches give players two shoes of various

collar heights, a mid-cut (MC) and a low-top (LT) volleyball shoe. Players then have the choice to wear either shoe while playing. More recently, volleyball players have been wearing MC that are a type of high collar shoes. The collar height on MC rises just superior to the talocrural joint and malleoli. With a higher collar height, the MC shoe has been introduced into the sport to restrict frontal-plane ankle motion as a method to combat another frequent injury, a lateral ankle sprain (Kaminski et al., 2019). The design of these higher collar shoes has been integrated into the sport to restrict frontal plane ankle motion and yet has also been shown to restrict the sagittal plane ankle range of motion, which has implications at the knee. Limited dorsiflexion range of motion has been associated to greater frontal plane knee excursion, peak knee abduction angles, peak external knee abduction moments, vertical and posterior ground reaction forces (GRF), knee anterior shear force, and smaller knee flexion angles (Fong et al., 2011b; Hoch et al., 2015; Schroeder et al., 2021). When increasing active dorsiflexion range of motion by 2° of motion, Schroeder et al. (2021) observed an increase knee flexion angle and decreased knee adduction angle when landing from a right-leg drop jump landing task. These aforementioned changes may increase the risk of athletes sustaining ACL injuries, as they have been identified as mechanisms of ACL ruptures (Hewett et al., 2005; Padua et al., 2015). Thus, the manipulation of sagittal plane range of motion at the ankle, specifically dorsiflexion range of motion, is important when considering the design and use of

higher collar shoes. High-top football cleats and basketball shoes, for example, have been shown to reduce dorsiflexion range of motion (Daack & Senchina, 2014b; Lam et al., 2015; Li et al., 2013; Tang et al., 2020). Very few studies have investigated the effect of higher collar shoes on both ankle and knee and mechanics. Vanwanseele et al. (2014), for example, investigated both ankle and knee joint biomechanics during landing and reported increase in peak tibial internal rotation moment while wearing a high-top basketball shoe compared to a LT netball shoe. Consequently, more research of the prevalent MC worn by volleyball players on knee landing mechanics is warranted.

Limb dominance is another risk factor for ACL injuries that have been identified in volleyball players. Kalata et al. (2020) identified volleyball as a sport comprised of an unbalanced amounts of unilateral loading from both take-off and landing movements between limbs. Observations in greater peak vertical loading, peak lateral force and rate of lateral loading which are higher pre-disposition to ACL injury have been seen in when volleyball players land unilaterally compared to on both limbs (Tillman, Criss, et al., 2004). This imbalance between limbs may lead to the development of asymmetries in muscular strength and landing movement strategies in volleyball players (Markou & Vagenas, 2006; Sinsurin et al., 2017). Limb dominance has been characterized as one limb having greater dynamic control to more effectively absorb the high impact forces during landing (Edwards et al., 2012; Ford et al., 2003b; Hewett et al., 2010). This discrepancy between limb control may place both limbs at risk for ACL injury, albeit for different reasons. For example, a higher dependence on the dominant limb to dissipate landing forces or the inability of the non-dominant limb to withstand large impact landing forces may influence the risk of injury at each limb (Ford et al., 2003b). The limb associated with greater injury risk in volleyball players, however, remains unclear. Avedesian et al. (2019) defined limb dominance subjectively as the leg participants would use to kick a soccer with and observed that female volleyball players landed with greater internal knee adduction moments on their non-dominant (NDL) limb. Sinsurin et al. (2017) defined the dominant limb as the limb that achieved a greater hopping distance in a single leg hop protocol and observed greater knee abduction angles and internal knee adduction moments when female volleyball players landed on the dominant limb (DL). Although similar populations were evaluated, and a similar unilateral landing task was used, the results were inconclusive as to which limb might be at greater risk for injury.

The interaction between shoe collar height and limb dominance, specifically in volleyball, and the effects on knee joint landing mechanics is unclear. The interaction between shoes and limbs is important as volleyball is a multidirectional sport that requires frequent unilateral landings on each limb (Lobiatti et al., 2010). It is important for volleyball coaches and players to assess the effect of both shoes on both limbs to better understand the impact of shoes on the entire lower extremity. Understanding the impact of shoes on the lower extremity while landing may aid in the decision by coaches and players in which footwear is more suitable to reduce injury risk while playing. Therefore, the purpose of this study was to determine the effects of MC and limb dominance on knee joint

biomechanics during a unilateral forward drop landing task in female collegiate volleyball players. We hypothesized that an interaction would be present between shoe and limb conditions. Specifically, we hypothesized that the DL in the MC shoe would have smaller sagittal plane knee joint angles and moments and larger frontal plane knee joint angles and moments than the NDL in the LT shoe. Additionally, as risk of ACL injury has been associated with both ankle mechanics and GRF (Hagins et al., 2007; Hoch et al., 2015; Malloy et al., 2015), we feel it pertinent to include these variables in our analysis. Thus, it was hypothesized that smaller sagittal plane and frontal plane ankle mechanics and greater vertical and posterior directed GRF would be observed when landing while wearing the MC. Previous research in female volleyball and netball players, and gymnasts found no significant differences between vertical GRF across the DL and NDL when performing a single-leg drop landing (Mokhtarzadeh et al., 2017). Additionally, horizontal GRF have been observed to remain similar across legs when performing a single-leg hop (Van der Harst et al., 2007). Therefore, we hypothesized all GRF would remain similar between limbs.

2. Methods

2.1. Participants

A convenience sample of one entire female National Collegiate Athletic Association (NCAA) Division-1 college volleyball team ($N = 18$) was recruited to attend one laboratory session. The inclusion criteria included collegiate female volleyball players who were current members of the same collegiate volleyball program. Exclusion criteria included participants who had any past or current musculoskeletal injury that the team physician deemed unsuitable to participate in the study (e.g., ACL reconstruction). One participant was excluded after not receiving clearance to participate from the team physician, thus a sample of 17 (height: 1.76 ± 0.06 m, mass: 72.29 ± 9.52 kg) athletes were used for our final analysis. The sample size was consistent with previous research assessing ankle dorsiflexion during landing (Fong et al., 2011b) and its impact at the knee (Schroeder et al., 2021). All participants provided informed consent approved by the University's Institutional Review Board and participated in the study.

2.2. Instrumentation

Lower extremity three-dimensional (3D) marker coordinate data were collected at 240 Hz using a 6-camera motion capture system (Qualisys, Gothenburg Sweden). GRF data were synchronously recorded using one in-ground force plate (1200 Hz, AMTI Inc., Watertown, MA). Single anatomic reflective anatomic markers were palpated and placed bilaterally on the acromion process, iliac crest, greater trochanter, medial and lateral femoral epicondyles, medial and lateral malleoli, distal end of the second toe, and the first and fifth metatarsal heads and used to define segments. Rigid thermoplastic segmental tracking clusters were placed on the trunk and pelvis, and bilaterally on the thighs, shanks, and heel and used to track



Figure 1. The adidas crazyflight volleyball shoes used in the current study. Mid-cut (left) and Low-Top (right) with the difference in collar height indicated by dashed lines.

segmental motion. A new static trial and musculoskeletal model were generated for each shoe condition.

2.3. Procedures

Participants completed five unilateral landing trials on both limbs in both a standard LT volleyball shoe and MC volleyball shoe (Crazy Flight, Adidas AG, Herzogenaurach, Germany, [Figure 1](#)). The order of the landing limb followed by the order of the shoe condition was randomized. The shoes had identical characteristics of a lightweight stretch mesh upper, a thermoplastic polyurethane yarn reinforced midfoot, a moulded sock liner, a boost midsole and a Top Grip rubber outsole, with the only difference being the height of the shoe collar. The DL was determined anecdotally by asking which leg the participants would kick a soccer ball with ([Avedesian et al., 2019](#); [Mercado-Palomino et al., 2021](#)).

The forward drop landing task began with the participant standing on a 30-cm tall box placed 40% of the participant's height away from the leading edge of the force plate ([Fong et al., 2011b](#); [Schroeder et al., 2021](#)). The participants were instructed to jump off the box horizontally, off one foot and land on the opposite foot on the force platform and maintain balance for 3 seconds. Prior to data collection, a familiarization period was given to ensure participants were comfortable with the task. The time of this period was dependent on when the participants felt comfortable with the task. Participants were given 30 seconds of rest between trials. After five successful trials were recorded for each limb in one shoe condition, a rest period of 3 min was given before repeating the procedures in the second shoe condition. A trial was deemed successful if the participant landed on the opposite foot on the force platform and were able to maintain balance for 3 seconds.

All variables were identified during the landing phase of the unilateral task. The landing phase was defined as initial foot contact with the force platform to peak knee flexion ([Garcia et al., 2022](#); [Schroeder et al., 2021](#); [Xu et al., 2020](#)). Initial contact was defined as the first time point vertical GRF reached the threshold of 10 N. Reduced sagittal plane motion and increased frontal plane motion in lower extremity mechanics at both initial contact and throughout landing have been proposed to elicit changes to ACL shear loads ([Markolf et al., 1995](#)) and have been targeted in training regiments to reduce injury risk ([Chappell & Limpisvasti, 2008](#); [Myer et al., 2006](#)). Thus, the variables of interest at the knee included both initial contact and peak knee joint

angles, and internal peak knee joint moments. Initial contact and peak ankle joint angles, internal peak ankle joint moments, and peak ground reaction forces were also variables of interest included for this study.

2.4. Data processing

Raw 3D marker coordinate data and GRF data were low-pass filtered using a fourth-order, zero lag, recursive Butterworth filter with a cut-off frequency of 10 Hz ([Movahed et al., 2019](#); [Xu et al., 2020](#)). A musculoskeletal model was created from the static calibration trial. The hip joint centre was placed 25% of the distance from the ipsilateral to contralateral greater trochanter markers ([Weinhandl & O'Connor, 2010](#)). The knee joint centres were defined as the midpoint between the medial and lateral femoral epicondyle markers ([Grood & Suntay, 1983](#)). The ankle joint centre was defined as the midpoint between the medial and lateral malleoli markers ([Wu et al., 2002](#)). Lower extremity joint angles were computed in the joint coordinate system ([Grood & Suntay, 1983](#)). Internal joint moments were computed with a Newton-Euler inverse dynamics approach and resolved in the joint coordinate system ([Grood & Suntay, 1983](#)). Positive rotations were defined as dorsiflexion and inversion at the ankle, and extension and adduction at the knee, and all computations were performed in Visual3d (Version 6, C-Motion, Inc., Germantown, MD, USA).

2.5. Statistical analysis

Descriptive statistics, including means and standard deviations from all five trials, were calculated for all dependent variables. A two-way [Limb (DL vs. NDL) $\times$ Shoe (MC vs LT)] repeated-measures ANOVA with a Bonferroni correction was performed to evaluate the impact of the shoe conditions on the GRF, knee and ankle biomechanical variables. Partial eta squared (η_p^2) was calculated as the effect size of the repeated measures ANOVA. Small, medium and large effect sizes corresponded to the values of 0.0099, 0.0588 and 0.1379, respectively ([Norouzzian & Plonsky, 2018](#); [Richardson, 2011](#)). If a significant interaction was found, post-hoc paired sample *t*-tests with Cohen's *d* effect sizes were used to determine statistically significant relationships with a Bonferroni correction to control for type I error rate. Cohen's *d* small, medium and large effect sizes corresponded to the values of 0.2, 0.5 and 0.8, respectively ([Rice & Harris, 2005](#)). Statistical significance was set at $\alpha=0.05$. All statistical analyses were performed using SPSS software (version 27, SPSS, Chicago, IL).

Table 1. Participant demographics presented as mean $\pm$ sd.

Age (yrs)	20.12 $\pm$ 1.32
Weight (kg)	72.29 $\pm$ 9.52
Height (cm)	175.53 $\pm$ 6.37
Shoe Size (US)	10.18 $\pm$ 1.26
Right Foot Dominant	N = 16
Left Foot Dominant	N = 1

3. Results

Participant demographics are reported in Table 1. In a standard US women's size 9 Adidas Crazyflight volleyball shoe, the MC collar height was 11 centimetres (cm), and the LT collar height was 9 cm (Figure 1). The mass of the MC and LT were measured to be 340.2 g and 289.17 g, respectively. A two-way repeated measures ANOVA did not reveal any statistically significant interactions between shoe and limb for any knee kinematic variables (Table 2). There was a significant main effect of shoe for peak dorsiflexion angle ($p = 0.038$), indicating greater peak dorsiflexion with LT (Table 2).

A significant interaction between shoe and limb was observed for internal peak knee flexion moment ($F(1, 17) = 4.620$, $p = 0.047$, Table 3). However, post-hoc analysis did not reveal any statistically

significant pairwise comparisons. A significant main effect of limb was found for internal peak knee abduction moment ($p = 0.014$, Table 3, Figure 2(b)), indicating greater peak knee abduction moment at the DL. A main effect of shoe was observed at the ankle, with greater internal peak plantarflexion moment in the LT ($p = 0.019$, Table 3, Figure 2(c)).

A significant interaction was found for peak lateral GRF ($F(1, 17) = 8.083$, $p = 0.012$). A post-hoc analysis revealed greater peak lateral GRF in the LT shoe compared to the MC for the NDL ($p = 0.003$, Cohen's $d = 0.851$, Figure 3(a)). A significant main effect of shoe was found for peak medial GRF ($p = 0.017$, Figure 3(a)), suggesting peak medially oriented GRF was greater in MC shoes.

4. Discussion

The purpose of this investigation was to determine the effects of shoe collar height and limb dominance on knee joint landing mechanics in female collegiate volleyball players. We hypothesized smaller knee sagittal plane knee kinematics and kinetics and greater knee frontal plane kinematics and kinetics when landing in the MC and on the DL. It

Table 2. Kinematic variables of the knee and ankle joints. Both peak angles and angles at initial contact (IC) are presented as mean $\pm$ sd for the non-dominant low-top shoe (ND LT), non-dominant mid-cut shoe (ND MC), dominant low-top shoe (D LT) and dominant mid-cut shoe (D MC). Interaction and main effect results are presented as p-value (η^2). **bold** represents statistical significance.

	ND LT	ND MC	D LT	D MC	Shoe	Limb	Interaction
Ankle							
Plantarflexion Angle	-34.9 $\pm$ 4.71	-34.1 $\pm$ 3.22	-32.7 $\pm$ 6.70	-33.5 $\pm$ 3.44	1.00 (0.000)	0.140 (0.131)	0.236 (0.087)
Dorsiflexion Angle	7.7 $\pm$ 5.41	6.1 $\pm$ 4.85	9.1 $\pm$ 4.44	7.2 $\pm$ 4.39	0.038 (0.242)	0.178 (0.110)	0.745 (0.007)
Inversion Angle	0.1 $\pm$ 5.31	0.9 $\pm$ 4.92	1.3 $\pm$ 5.28	2.3 $\pm$ 4.81	0.148 (0.126)	0.231 (0.088)	0.868 (0.002)
Eversion Angle	-9.8 $\pm$ 2.67	-8.8 $\pm$ 3.49	-10.5 $\pm$ 3.52	-11.0 $\pm$ 2.54	0.658 (0.013)	0.120 (0.145)	0.156 (0.121)
Sagittal Plane IC	-35.1 $\pm$ 4.69	-34.2 $\pm$ 3.28	-30.2 $\pm$ 12.99	-33.5 $\pm$ 3.44	0.474 (0.033)	0.081 (0.178)	0.156 (0.122)
Frontal Plane IC	-0.5 $\pm$ 5.73	0.4 $\pm$ 5.22	1.2 $\pm$ 5.35	2.2 $\pm$ 4.93	0.139 (0.131)	0.148 (0.126)	0.901 (0.001)
Knee							
Flexion Angle	-57.2 $\pm$ 8.23	-57.6 $\pm$ 8.87	-59.4 $\pm$ 7.65	-59.5 $\pm$ 7.69	0.848 (0.002)	0.227 (0.090)	0.832 (0.003)
Extension Angle	-14.3 $\pm$ 5.35	-14.1 $\pm$ 5.27	-14.3 $\pm$ 3.95	-14.2 $\pm$ 4.79	0.703 (0.009)	0.991 (0.000)	0.843 (0.003)
Abduction Angle	-0.4 $\pm$ 3.21	-1.0 $\pm$ 3.18	-1.9 $\pm$ 4.02	-1.2 $\pm$ 2.82	0.913 (0.001)	0.330 (0.059)	0.083 (0.176)
Adduction Angle	4.9 $\pm$ 3.02	4.7 $\pm$ 2.93	4.8 $\pm$ 3.27	5.5 $\pm$ 3.25	0.682 (0.011)	0.626 (0.015)	0.243 (0.084)
Sagittal Plane IC	-14.3 $\pm$ 5.32	-14.0 $\pm$ 5.25	-14.3 $\pm$ 3.95	-14.2 $\pm$ 4.64	0.709 (0.009)	0.928 (0.001)	0.672 (0.011)
Frontal Plane IC	2.1 $\pm$ 1.83	2.0 $\pm$ 2.33	1.8 $\pm$ 2.51	2.0 $\pm$ 2.48	0.821 (0.003)	0.767 (0.006)	0.622 (0.016)

Table 3. Ground reaction forces and ankle and knee joint kinetics. Ground reaction forces are presented as mean $\pm$ sd in % body weight (BW) and internal peak joint moments are presented as mean $\pm$ sd in % body mass (BM) for the non-dominant low-top shoe (ND LT), non-dominant mid-cut shoe (ND MC), dominant low-top shoe (D LT) and dominant mid-cut shoe (D MC). Interaction and main effect results are presented as p-value (η^2). **Bold** represents statistical significance.

	ND LT	ND MC	D LT	D MC	Shoe	Limb	Interaction
Mediolateral							
Max	0.0102 $\pm$ 0.02 ^x	0.0003 $\pm$ 0.02	0.0068 $\pm$ 0.03	0.0171 $\pm$ 0.03	0.968 (0.000)	0.309 (0.064)	0.012 (0.336)
Min	-0.1232 $\pm$ 0.03	-0.1396 $\pm$ 0.04	-0.1216 $\pm$ 0.03	-0.1219 $\pm$ 0.03	0.017 (0.309)	0.204 (0.099)	0.196 (0.102)
Anteroposterior							
Max	-0.0815 $\pm$ 0.01	-0.0917 $\pm$ 0.02	-0.0766 $\pm$ 0.03	-0.0816 $\pm$ 0.02	0.083 (0.176)	0.097 (0.163)	0.625 (0.015)
Vertical							
Max	2.8442 $\pm$ 0.36	2.8050 $\pm$ 0.32	2.7688 $\pm$ 0.38	2.7653 $\pm$ 0.37	0.563 (0.021)	0.155 (0.122)	0.456 (0.035)
Ankle							
Dorsiflexion Moment	-0.0 $\pm$ 0.02	-0.0 $\pm$ 0.06	0.0 $\pm$ 0.17	-0.2 $\pm$ 0.05	0.395 (0.046)	0.142 (0.130)	0.138 (0.132)
Plantarflexion Moment	-1.8 $\pm$ 0.34	-1.6 $\pm$ 0.24	-1.7 $\pm$ 0.45	-1.6 $\pm$ 0.34	0.019 (0.298)	0.689 (0.010)	0.079 (0.180)
Eversion Moment	-0.2 $\pm$ 0.12	-2 $\pm$ 0.13	-0.1 $\pm$ 0.05	-0.1 $\pm$ 0.09	0.359 (0.053)	0.121 (0.144)	0.473 (0.033)
Inversion Moment	0.1 $\pm$ 0.12	0.1 $\pm$ 0.13	0.2 $\pm$ 0.07	0.1 $\pm$ 0.09	0.333 (0.059)	0.150 (0.125)	0.469 (0.033)
Knee							
Flexion Moment	-0.2 $\pm$ 0.09	-0.1 $\pm$ 0.15	-0.1 $\pm$ 0.12	-0.2 $\pm$ 0.10	0.190 (0.105)	0.520 (0.026)	0.047 (0.224)
Extension Moment	3.1 $\pm$ 0.53	3.1 $\pm$ 0.46	3.2 $\pm$ 0.47	3.2 $\pm$ 0.47	0.277 (0.073)	0.544 (0.023)	0.759 (0.006)
Adduction Moment	0.1 $\pm$ 0.05	0.1 $\pm$ 0.06	0.0 $\pm$ 0.06	0.1 $\pm$ 0.10	0.459 (0.035)	0.592 (0.018)	0.442 (0.037)
Abduction Moment	-0.9 $\pm$ 0.24	-0.9 $\pm$ 0.28	-1.0 $\pm$ 0.25	-1.1 $\pm$ 0.30	0.280 (0.072)	0.014 (0.323)	0.068 (0.193)

^x = Statistically significant difference from the ND MC.

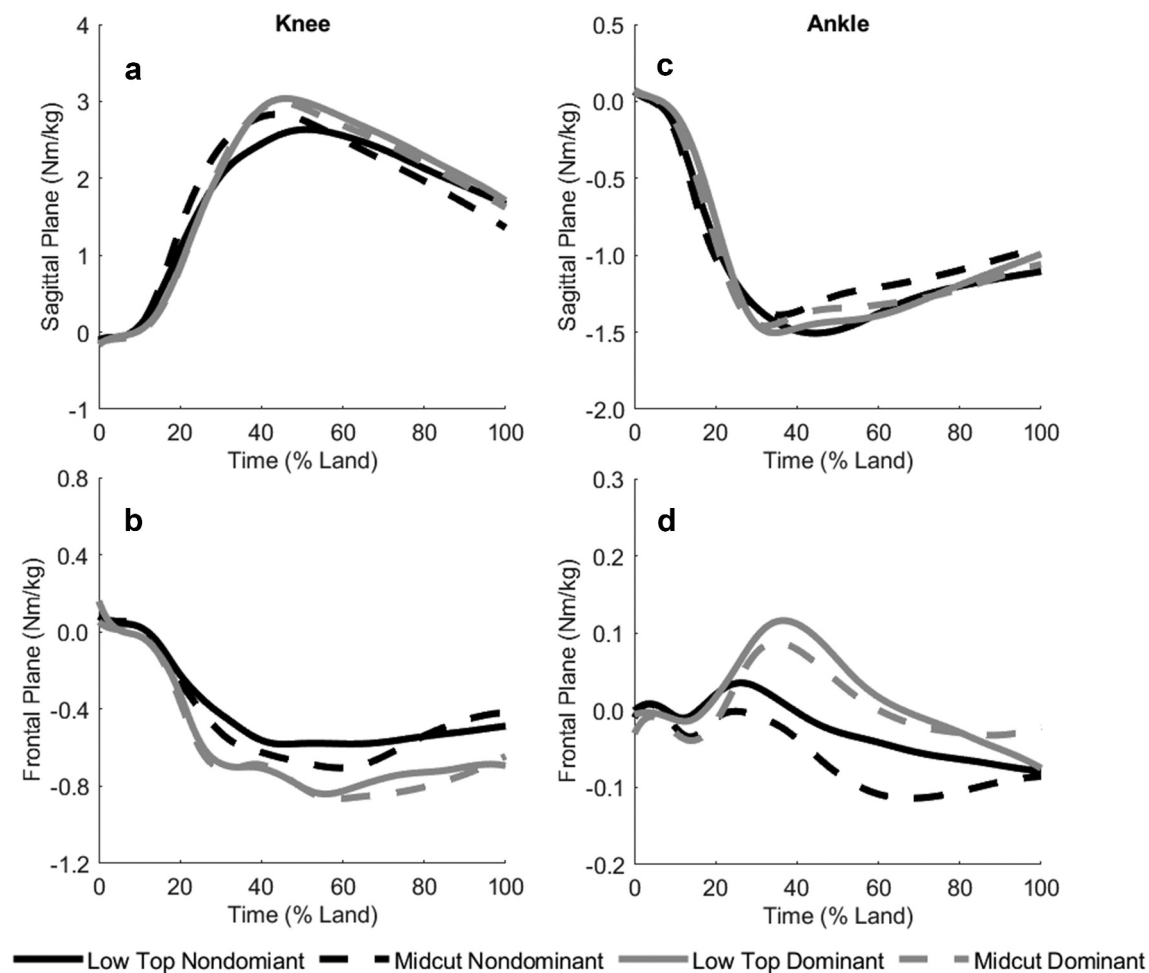


Figure 2. Knee and ankle joint moments during landing between shoe and limb. Sagittal plane (A) and frontal plane (B) internal peak knee joint moments. Sagittal (C) and frontal (D) internal peak ankle joint moments.

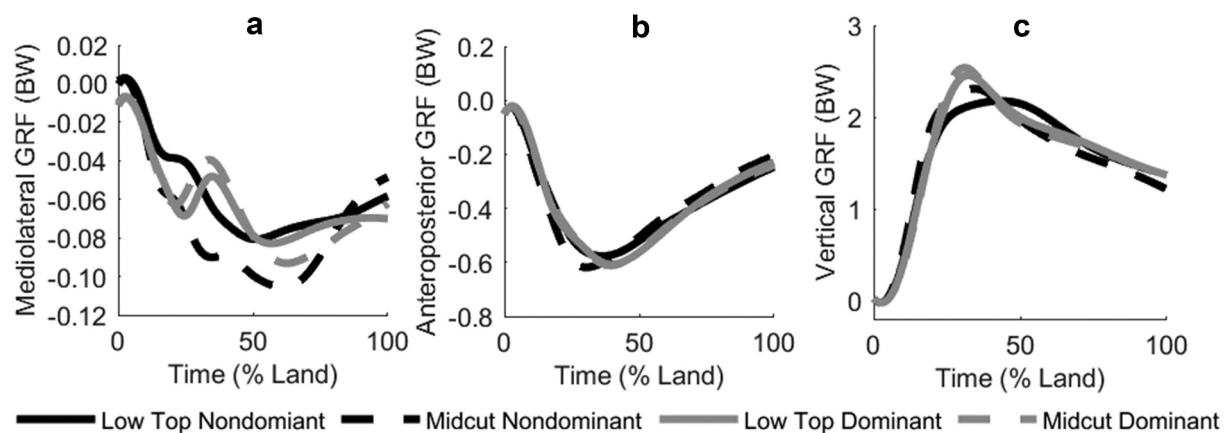


Figure 3. Mediolateral (A), anteroposterior (B) and vertical (C) ground reaction forces during landing.

was also hypothesized that participants would land with greater vertical and posterior GRF in the MC but with GRF remaining similar between limbs. Our hypotheses regarding the interaction between shoe and limb were not supported as only mediolateral GRF and peak knee flexion moment

were significant. The paired samples *t*-test did not identify any significant differences in peak knee flexion moment while the location of statistical significance between mediolateral GRF was found between the NDL MC and the NDL LT which will be later discussed.

4.1. Shoe collar height

Post-hoc tests revealed a significant difference in mediolateral GRF between the two shoe conditions. We found that participants landed with greater medially directed GRF while landing in the MC (Figure 3). Although larger medial GRF were observed while wearing the MC shoe, the magnitude of peak medial GRF between shoe collar heights does not seem to be large enough to affect the frontal plane knee and ankle kinematics and kinetics when landing. Contrary to our hypotheses, no statistically significant differences were found for initial contact and peak joint angles or peak joint moments at the knee between shoe conditions (Tables 2, 3). This finding was surprising to us because we found the MC shoe reduced the peak dorsiflexion angle (Table 2) and reduced the internal peak plantarflexion moment compared to the LT (Figure 2(c)). With the previously established association between reduced passive ankle dorsiflexion range of motion and increased frontal plane knee angles and moments, we anticipated changes in knee mechanics (Hagins et al., 2007; Malloy et al., 2015; Sigward et al., 2008). However, the changes at the ankle in the MC were not accompanied by any differences in knee kinematics or kinetics. Although the paired samples *t*-test indicated a statistically significant difference in peak dorsiflexion angle between shoes, this difference may not be functionally significant, suggesting that the design of the MC may not be restrictive enough at the ankle to in return, induce changes at the knee. Between the MC and LT, there was a 2 cm difference in collar height. Previously, Liu et al. (2017) reported a 5.7 cm difference between a high-top and LT Li Ning basketball shoe and Yang et al. (2017) reported a 4.3 cm difference between the high-top and LT Nike basketball shoe. This additional collar height affords the ability to tie the laces above the malleoli which provides additional restriction of the ankle (Rowson et al., 2010). Potentially, the structural design differences between an MC and a true high top may be large enough to change how each type of shoe impacts lower extremity mechanics. However, it is important to note, these mentioned studies did not report any knee mechanics, making the impact of the higher collar shoes unclear. Vanwanseele et al. (2014) compared a high-top basketball shoe to a LT netball shoe at both the ankle and knee during landing and reported increased peak knee internal rotation moment in high-top shoes. Vanwanseele et al. (2014) used two different styles and sport shoes in a population of netball players that were unfamiliar with playing in a high-top, making their results hard to compare with the present study. The properties of the increased collar height of the MC in this study were not sufficiently different from the LT to impact knee mechanics while landing.

Smaller dorsiflexion range of motion has been associated with greater vertical and posterior GRF (Fong et al., 2011b; Hagins et al., 2007; Hoch et al., 2015). Based on this association, it was anticipated that the vertical and posteriorly directed GRF would be greater while landing in the MC compared to the LT. Contrary to our hypotheses, no changes were observed in the vertical and posterior GRF between shoes. These results are consistent with Vanwanseele et al. (2014), who observed comparable peak vertical GRF between a high top and an LT. However, previous research has found moderate associations

between decreased passive ankle dorsiflexion range of motion in a LT, increased vertical GRF (Fong et al., 2011b; Hoch et al., 2015), and increased posterior GRF (Fong et al., 2011b; Hoch et al., 2015). In the study conducted by Hoch et al. (2015), the participants landed with initial contact angles of 26.24° of plantar flexion and 13.21° of knee flexion while in the current study, the participants landed with initial contact angles varying between 30–36° of plantarflexion and 14° of knee flexion (Table 2). Greater sagittal plane excursion during the landing phase attenuates forces throughout the lower extremity over a greater duration of time (Fong et al., 2011b; Hoch et al., 2015). Given that the volleyball athletes in the current study landed in a position of increased plantarflexion and knee flexion, they may have adapted different landing techniques in the sagittal plane at both the knee and ankle which may have contributed to similar vertical and posterior GRF across shoe conditions by increasing the duration of the landing phase. Caution is warranted when comparing the results of our studies to others as different marker placements and tasks may have impacted study results. Nonetheless, further investigation is warranted.

For the implication of lower extremity injury risk in volleyball, the greater mediolateral GRF and smaller peak dorsiflexion angle did not impact frontal plane knee mechanics that had been previously associated with ACL injuries. As previously noted, the MC have been implemented into volleyball to restrict frontal plane ankle motion, yet there were no observed restrictions of frontal plane ankle motion that are associated with lateral ankle sprains. In the decision-making process for both players and coaches, the MC or LT may be worn without significantly affecting landing mechanics at the knee joint.

4.2. Limb dominance

While wearing the MC in the NDL, the mediolateral GRF was directed more medial than wearing the LT in the NDL (Figure 3(a), Table 3). Qualitatively, mediolateral GRF was directed more medial in the LT than the MC in the DL (Figure 3(a)). Depending on the limb, the direction of the GRF differed between shoe conditions. This suggests that force-production strategies between dominant and NDL may play a greater role in GRF production than shoe collar height. In the current study, the NDL had greater medially directed GRF (Figure 3(a)) and smaller internal peak knee abduction moments (Figure 2(b)) compared to the DL when landing. The medially directed GRF in the NDL may be a positive finding in our sample of volleyball athletes, as greater medial GRF has been associated with reduction in frontal plane angles and moments at the knee associated with ACL injuries (Seymore et al., 2019; Ueno et al., 2020).

4.3. Limitations

Results of this study need to be considered within in the context of several limitations. First, leg dominance was measured subjectively. Although some other studies have reported volleyball players' leg dominance in the same way, other methods have been utilized such as strength measures. The inconsistent methodology in defining the dominant limb makes the comparing results difficult. Devetag et al. (2018) observed that the

highest level of female professional volleyball players more frequently ruptured their ACL on their left leg. Instead of leg dominance, reporting right or left leg may be a better solution to reduce ambiguity of which leg is potentially at greater risk of injury in sports. The forward unilateral landing task was utilized in this study as it has been validated as a clinical assessment of landing biomechanics (Padua et al., 2015) and better represents the multiplanar nature of sports (Cruz et al., 2013). However, this task does not replicate the multidirectional three-step sequence volleyball players use to jump and land. Second, the participants in this study were a part of a convenience sample from one collegiate volleyball program. This population only represents a small portion of Division I NCAA volleyball players, and the participants were not classified by their position. Additionally, the foot type of each participant was not considered in this study which may influence landing techniques. Lastly, the shoes provided by the participants themselves in the study may be a limitation. Shoes worn by the participants in this study were the team shoes given out at the start of the volleyball season. Although each shoe was the same age, make and model, the wear and degradation of the shoes were not quantified and may have impacted landing mechanics.

5. Conclusion

The reductions in peak dorsiflexion angle and internal peak plantarflexion moment observed while wearing the MC were not sufficiently influential to increase the frontal plane knee kinematics and kinetics associated with ACL injuries. We recommend that volleyball coaches and athletic training staff consider these results of this study when making the decision of footwear to improve performance and reduce injury risk. Collegiate volleyball players demonstrated greater laterally directed GRF and internal knee abduction moments when landing on their DL, but further conclusive research is warranted to better assess which limb is at greater risk of various knee injuries. Training regiments should focus on improving knee joint landing mechanics in the DL of volleyball players to reduce injury risk at the knee while playing.

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References

Avedesian, J. M., Judge, L. W., Wang, H., & Dickin, D. C. (2019). Kinetic analysis of unilateral landings in female volleyball players after a dynamic and combined dynamic-static warm-up. *The Journal of Strength & Conditioning Research*, 33(6), 1524–1533. <https://doi.org/10.1519/JSC.0000000000002736>

Bisseling, R. W., Hof, A. L., Bredeweg, S. W., Zwerver, J., & Mulder, T. (2008). Are the take-off and landing phase dynamics of the volleyball spike jump related to patellar tendinopathy? *British Journal of Sports Medicine*, 42(6), 483–489. <https://doi.org/10.1136/bjsm.2007.044057>

Chandran, A., Morris, S. N., Lempke, L. B., Boltz, A. J., Robison, H. J., & Collins, C. L. (2021). Epidemiology of Injuries in National Collegiate Athletic Association Women's Volleyball: 2014–2015 through 2018–2019. *Journal of Athletic Training*, 56(7), 666–673. <https://doi.org/10.4085/1062-6050-679-20>

Chappell, J. D., & Limpisvasti, O. (2008). Effect of a neuromuscular training program on the kinetics and kinematics of jumping tasks. *The American Journal of Sports Medicine*, 36(6), 1081–1086. <https://doi.org/10.1177/0363546508314425>

Cronin, A. (2001). *Effect of selected ankle supports on resistance to inversion force and range of motion*. University of Ottawa (Canada). <https://doi.org/10.20381/ruor-16154>

Cruz, A., Bell, D., McGrath, M., Blackburn, T., Padua, D., & Herman, D. (2013). The effects of three jump landing tasks on kinetic and kinematic measures: Implications for ACL injury research. *Research in Sports Medicine*, 21, 330–342. <https://doi.org/10.1080/15438627.2013.825798>

Daack, C. W., & Senchina, D. S. (2014a). A field study of low-top vs. mid-top vs. high-top American football cleats. *Sports*, 2(4), 85–98. <https://doi.org/10.3390/sports2040085>

Daack, C. W., & Senchina, D. S. (2014b). A field study of low-top vs. mid-top vs. high-top American football cleats. *Sports*, 2, 85–98. <https://doi.org/10.3390/sports2040085>

Devetag, F., Mazzilli, M., Benis, R., La Torre, A., & Bonato, M. (2018). Anterior cruciate ligament injury profile in Italian serie A1-A2 women's volleyball league. *The Journal of Sports Medicine and Physical Fitness*, 58, 92–97. <https://doi.org/10.23736/S0022-4707.16.06731-1>

Edwards, S., Steele, J. R., Cook, J. L., Purdam, C. R., & McGhee, D. E. (2012). Lower limb movement symmetry cannot be assumed when investigating the stop-jump landing. *Medicine and Science in Sports and Exercise*, 44(6), 1123–1130. <https://doi.org/10.1249/mss.0b013e31824299c3>

Ferretti, A., Papandrea, P., Contedua, F., & Mariani, P. P. (1992). Knee ligament injuries in volleyball players. *The American Journal of Sports Medicine*, 20(2), 203–207. <https://doi.org/10.1177/036354659202000219>

Fong, C.-M., Blackburn, J. T., Norcross, M. F., McGrath, M., & Padua, D. A. (2011a). Ankle-dorsiflexion range of motion and landing biomechanics. *Journal of Athletic Training*, 46, 5–10. <https://doi.org/10.4085/1062-6050-46.1.5>

Fong, C.-M., Blackburn, J. T., Norcross, M. F., McGrath, M., & Padua, D. A. (2011b). Ankle-dorsiflexion range of motion and landing biomechanics. *Journal of Athletic Training*, 46, 5–10. <https://doi.org/10.4085/1062-6050-46.1.5>

Ford, K. R., Myer, G. D., & Hewett, T. E. (2003a). Valgus knee motion during landing in high school female and male basketball players. *Medicine & Science in Sports & Exercise*, 35(10), 1745–1750. <https://doi.org/10.1249/01.MSS.0000089346.85744.D9>

Ford, K. R., Myer, G. D., & Hewett, T. E. (2003b). Valgus knee motion during landing in high school female and male basketball players. *Medicine & Science in Sports & Exercise*, 35(10), 1745–1750. <https://doi.org/10.1249/01.MSS.0000089346.85744.D9>

Garcia, S., Delattre, N., Berton, E., Divrechy, G., & Rao, G. (2022). Comparison of landing kinematics and kinetics between experienced and novice volleyball players during block and spike jumps. *BMC Sports Science, Medicine and Rehabilitation*, 14(1), 105. <https://doi.org/10.1186/s13102-022-00496-0>

Grood, E. S., & Suntay, W. J. (1983). A joint coordinate system for the clinical description of three-dimensional motions: Application to the knee. *Journal of Biomechanical Engineering*, 105(2), 136–144. <https://doi.org/10.1115/1.3138397>

Hagins, M., Pappas, E., Kremenich, I., Orishimo, K. F., & Rundle, A. (2007). The effect of an inclined landing surface on biomechanical variables during a jumping task. *Clinical Biomechanics*, 22(9), 1030–1036. <https://doi.org/10.1016/j.clinbiomech.2007.07.012>

Hewett, T. E., Ford, K. R., Hoogenboom, B. J., & Myer, G. D. (2010). Understanding and preventing acl injuries: Current biomechanical and epidemiologic considerations-update 2010. *North American Journal of Sports Physical Therapy: NAJSPT*, 5, 234.

- Hewett, T. E., Myer, G. D., Ford, K. R., Heidt Jr, R. S., Colosimo, A. J., McLean, S. G., Van den Bogert, A. J., Paterno, M. V., & Succop, P. (2005). Biomechanical measures of neuromuscular control and valgus loading of the knee predict anterior cruciate ligament injury risk in female athletes: A prospective study. *The American Journal of Sports Medicine*, 33(4), 492–501. <https://doi.org/10.1177/0363546504269591>
- Hoch, M. C., Farwell, K. E., Gaven, S. L., & Weinhandl, J. T. (2015). Weight-bearing dorsiflexion range of motion and landing biomechanics in individuals with chronic ankle instability. *Journal of Athletic Training*, 50(8), 833–839. <https://doi.org/10.4085/1062-6050-50.5.07>
- Kalata, M., Maly, T., Hank, M., Michalek, J., Bujnovsky, D., Kunzmann, E., & Zahalka, F. (2020). Unilateral and bilateral strength asymmetry among young elite athletes of various sports. *Medicina*, 56(12), 683. <https://doi.org/10.3390/medicina56120683>
- Kaminski, T. W., Needle, A. R., & Delahunt, E. (2019). Prevention of lateral ankle sprains. *Journal of Athletic Training*, 54(6), 650–661. <https://doi.org/10.4085/1062-6050-487-17>
- Lam, G. W. K., Park, E. J., Lee, K.-K., & Cheung, J. T.-M. (2015). Shoe collar height effect on athletic performance, ankle joint kinematics and kinetics during unanticipated maximum-effort side-cutting performance. *Journal of Sports Sciences*, 33(16), 1738–1749. <https://doi.org/10.1080/02640414.2015.1011206>
- Li, L., Liu, Y., He, J., & Fu, W. (2013). *Effects of shoe collar height on sagittal ankle rom, kinetics and power output during single-leg and double-leg jumps*. ISBS-Conference Proceedings Archive.
- Liu, H., Wu, Z., & Lam, W.-K. (2017). Collar height and heel counter-stiffness for ankle stability and athletic performance in basketball. *Research in Sports Medicine*, 25(2), 209–218.
- Lobiatti, R., Coleman, S., Pizzichillo, E., & Merni, F. (2010). Landing techniques in volleyball. *Journal of Sports Sciences*, 28(13), 1469–1476. <https://doi.org/10.1080/02640414.2010.514278>
- Malloy, P., Morgan, A., Meinerz, C., Geiser, C., & Kipp, K. (2015). The association of dorsiflexion flexibility on knee kinematics and kinetics during a drop vertical jump in healthy female athletes. *Knee Surgery, Sports Traumatology, Arthroscopy*, 23(12), 3550–3555. <https://doi.org/10.1007/s00167-014-3222-z>
- Markolf, K. L., Burchfield, D. M., Shapiro, M. M., Shepard, M. F., Finerman, G. A. M., & Slauterbeck, J. L. (1995). Combined knee loading states that generate high anterior cruciate ligament forces. *Journal of Orthopaedic Research*, 13(6), 930–935. <https://doi.org/10.1002/jor.1100130618>
- Markou, S., & Vagenas, G. (2006). Multivariate isokinetic asymmetry of the knee and shoulder in elite volleyball players. *European Journal of Sport Science*, 6(1), 71–80. <https://doi.org/10.1080/17461390500533147>
- Mercado-Palomino, E., Aragón-Royón, F., Richards, J., Benítez, J. M., & Ureña Espa, A. (2021). The influence of limb role, direction of movement and limb dominance on movement strategies during block jump-landings in volleyball. *Scientific Reports*, 11(1), 1–11. <https://doi.org/10.1038/s41598-021-03106-0>
- Mokhtarzadeh, H., Ewing, K., Janssen, I., Yeow, C.-H., Brown, N., & Lee, P. V. S. (2017). The effect of leg dominance and landing height on ACL loading among female athletes. *Journal of Biomechanics*, 60, 181–187. <https://doi.org/10.1016/j.jbiomech.2017.06.033>
- Movahed, M., Salavati, M., Sheikhhoseini, R., Arab, A. M., & O'Sullivan, K. (2019). Single leg landing kinematics in volleyball athletes: A comparison between athletes with and without active extension low back pain. *Journal of Bodywork and Movement Therapies*, 23(4), 924–929. <https://doi.org/10.1016/j.jbmt.2019.01.012>
- Myer, G. D., Ford, K. R., McLean, S. G., & Hewett, T. E. (2006). The effects of plyometric versus dynamic stabilization and balance training on lower extremity biomechanics. *The American Journal of Sports Medicine*, 34(3), 445–455. <https://doi.org/10.1177/0363546505281241>
- Norouzi, R., & Plonsky, L. (2018). Eta and partial eta squared in L2 research: A cautionary review and guide to more appropriate usage. *Second Language Research*, 34(2), 257–271. <https://doi.org/10.1177/02676583>
- Padua, D. A., DiStefano, L. J., Beutler, A. I., De La Motte, S. J., DiStefano, M. J., & Marshall, S. W. (2015). The landing error scoring system as a screening tool for an anterior cruciate ligament injury-prevention program in elite youth soccer athletes. *Journal of Athletic Training*, 50(6), 589–595. <https://doi.org/10.4085/1062-6050-50.1.10>
- Rice, M. E., & Harris, G. T. (2005). Comparing effect sizes in follow-up studies: ROC area, Cohen's d, and r. *Law and Human Behavior*, 29(5), 615–620. <https://doi.org/10.1007/s10979-005-6832-7>
- Richardson, J. T. E. (2011). Eta squared and partial eta squared as measures of effect size in educational research. *Educational Research Review*, 6(2), 135–147. <https://doi.org/10.1016/j.edurev.2010.12.001>
- Rowson, S., McNally, C., & Duma, S. M. (2010). Can footwear affect achilles tendon loading? *Clinical Journal of Sport Medicine*, 20(5), 344–349. <https://doi.org/10.1097/JSM.0b013e3181ed7e50>
- Schroeder, L. E., Tatarski, R. L., & Weinhandl, J. T. (2021). Increased ankle range of motion reduces knee loads during landing in healthy adults. *Journal of Applied Biomechanics*, 37(4), 333–342. <https://doi.org/10.1123/jab.2020-0281>
- Seymore, K. D., Fain, A. C., Lobb, N. J., Brown, T. N., & Barbosa, T. M. (2019). Sex and limb impact biomechanics associated with risk of injury during drop landing with body borne load. *Public Library of Science ONE*, 14(2), e0211129. <https://doi.org/10.1371/journal.pone.0211129>
- Sigward, S. M., Ota, S., & Powers, C. M. (2008). Predictors of frontal plane knee excursion during a drop land in young female soccer players. *Journal of Orthopaedic & Sports Physical Therapy*, 38(11), 661–667. <https://doi.org/10.2519/jospt.2008.2695>
- Sinsurin, K., Srisangboriboon, S., & Vachalathiti, R. (2017). Side-to-side differences in lower extremity biomechanics during multi-directional jump landing in volleyball athletes. *European Journal of Sport Science*, 17(6), 699–709. <https://doi.org/10.1080/17461391.2017.1308560>
- Tang, Y., Wang, Z., Zhang, Y., Zhang, S., Wei, S., Pan, J., & Liu, Y. (2020). Effect of football shoe collar type on ankle biomechanics and dynamic stability during anterior and lateral single-leg jump landings. *Applied Sciences*, 10(10), 3362. <https://doi.org/10.3390/app10103362>
- Tillman, M. D., Criss, R. M., Brunt, D., & Hass, C. J. (2004). Landing constraints influence ground reaction forces and lower extremity EMG in female volleyball players. *Journal of Applied Biomechanics*, 20(1), 38–50. <https://doi.org/10.1123/jab.20.1.38>
- Tillman, M. D., Hass, C. J., Brunt, D., & Bennett, G. R. (2004). Jumping and landing techniques in elite women's volleyball. *Journal of Sports Science & Medicine*, 3, 30.
- Ueno, R., Navacchia, A., DiCesare, C. A., Ford, K. R., Myer, G. D., Ishida, T., Tohyama, H., & Hewett, T. E. (2020). Knee abduction moment is predicted by lower gluteus medius force and larger vertical and lateral ground reaction forces during drop vertical jump in female athletes. *Journal of Biomechanics*, 103, 109669. <https://doi.org/10.1016/j.jbiomech.2020.109669>
- Van der Harst, J. J., Gokeler, A., & Hof, A. L. (2007). Leg kinematics and kinetics in landing from a single-leg hop for distance. A comparison between dominant and non-dominant leg. *Clinical Biomechanics*, 22(6), 674–680. <https://doi.org/10.1016/j.clinbiomech.2007.02.007>
- Vanwanseele, B., Stuelcken, M., Greene, A., & Smith, R. (2014). The effect of external ankle support on knee and ankle joint movement and loading in netball players. *Journal of Science & Medicine in Sport*, 17, 511–515. <https://doi.org/10.1016/j.jsams.2013.07.009>
- Weinhandl, J. T., & O'Connor, K. M. (2010). Assessment of a greater trochanter-based method of locating the hip joint center. *Journal of Biomechanics*, 43(13), 2633–2636. <https://doi.org/10.1016/j.jbiomech.2010.05.023>
- Wu, G., Siegler, S., Allard, P., Kirtley, C., Leardini, A., Rosenbaum, D., Whittle, M., D D'Lima, D., Cristofolini, L., & Witte, H. (2002). ISB recommendation on definitions of joint coordinate system of various joints for the reporting of human joint motion—part I: Ankle, hip, and spine. *Journal of Biomechanics*, 35, 543–548. [https://doi.org/10.1016/S0021-9290\(01\)00222-6](https://doi.org/10.1016/S0021-9290(01)00222-6)
- Xu, D., Jiang, X., Cen, X., Baker, J. S., & Gu, Y. (2020). Single-leg landings following a volleyball spike may increase the risk of anterior cruciate ligament injury more than landing on both-legs. *Applied Sciences*, 11(1), 130. <https://doi.org/10.3390/app11010130>
- Yang, Y., Fang, Y., Zhang, X., He, J., & Fu, W. (2017). Does shoe collar height influence ankle joint kinematics and kinetics in sagittal plane maneuvers? *Journal of Sports Science and Medicine*, 16(4), 543–550.